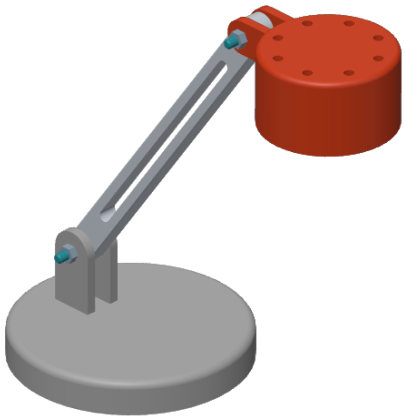
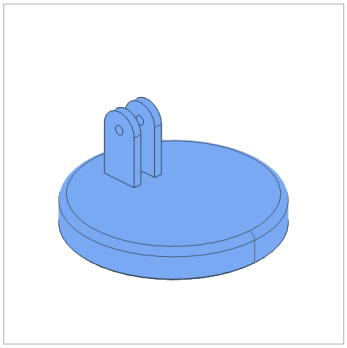
DESK LAMP ASSEMBLY

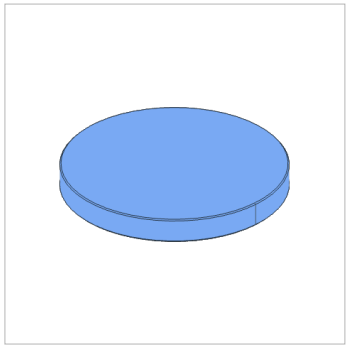
Putting one desk lamp together from its printed parts, its steel weight, the LED module and its screws.



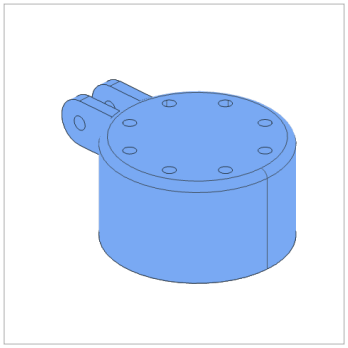
What you need



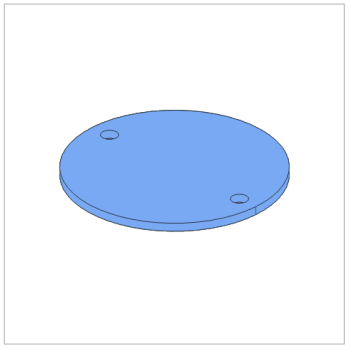
×1 base



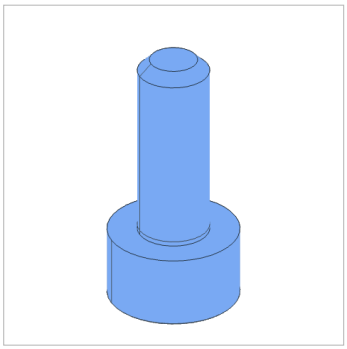
×1 weight



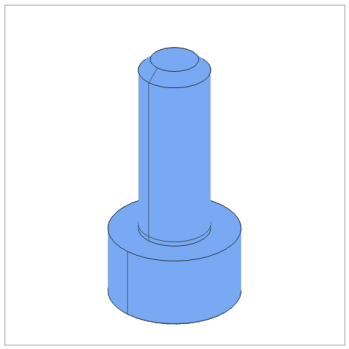
×1 head



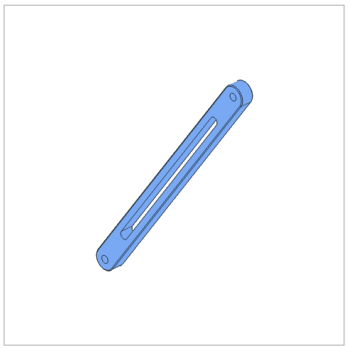
×1 Led module



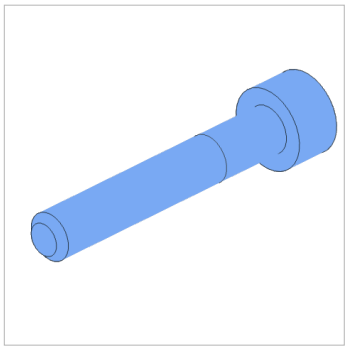
×1 M3x8 screw:left



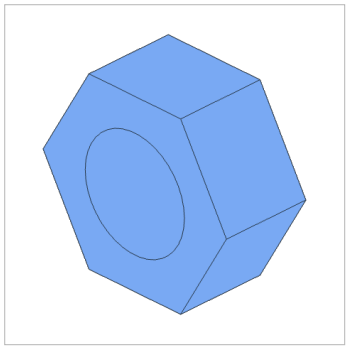
×1 M3x8 screw:right



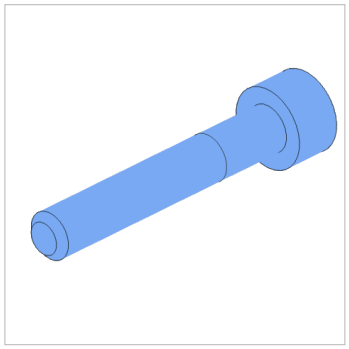
×1 arm



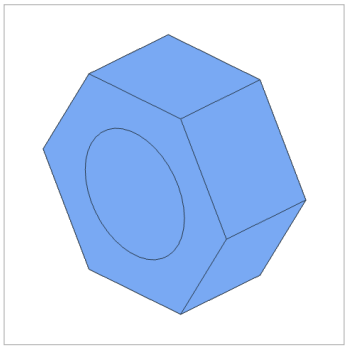
×1 M5x30 screw:base hinge



×1 M5 locknut:base hinge



×1 M5x30 screw:head hinge



×1 M5 locknut:head hinge

Also: ISO 4762 M3 x 8 socket head screw, qty 2, M5 nylon insert locknut on the M5 x 30 hinge screw, qty 1

Tools: 4 mm hex key, 8 mm spanner, 2.5 mm hex key

Still to decide

Step 2: adhesive not decided yet

Step 5: torque not decided yet

Step 5: source of 2.5 mm not decided yet

Step 8: torque not decided yet

Step 11: torque not decided yet

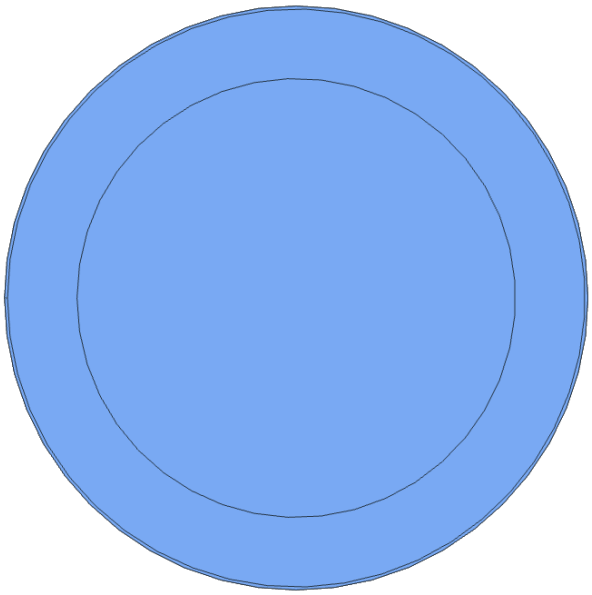
Step 12: source of 10 degrees not decided yet

Step 12: source of 10 degrees, 10 degrees not decided yet

Step 12: source of 55 degrees, 45 degrees, 175 degrees not decided yet

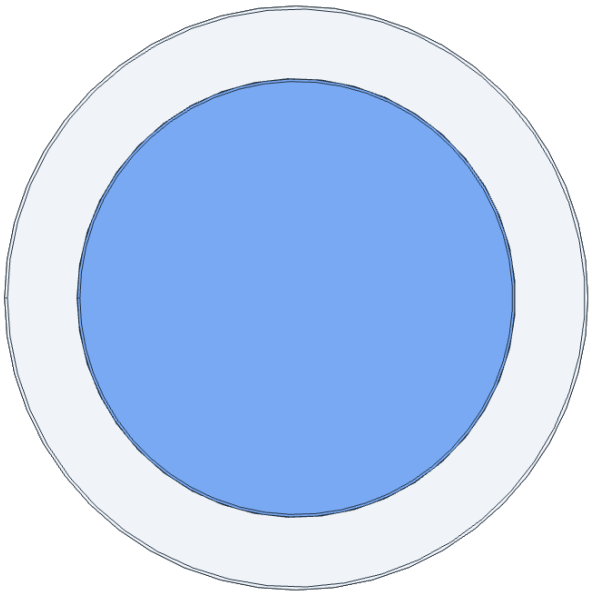
Blue shows what you add in each step.

1 Lay the base upside down on the bench



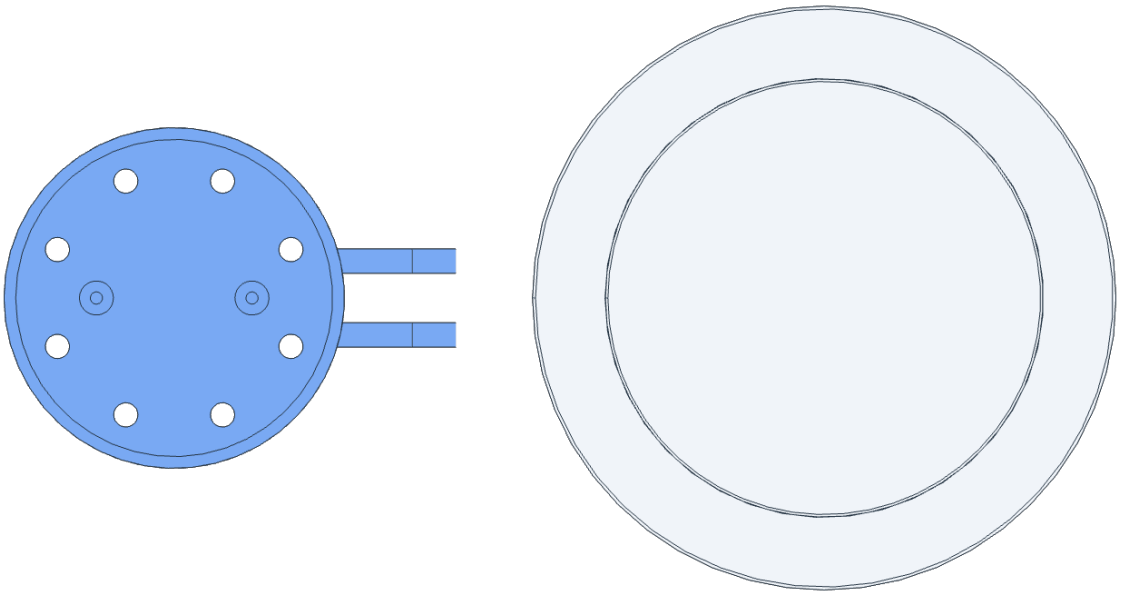
You need: ×1 base

2 Bond the steel weight into the base's pocket, flush with the underside



You need: ×1 weight
The purple outline shows where.
Adhesive: not decided yet
Check: The weight sits flat on the pocket's ceiling, its underside flush with the base's
Write down: flush, yes or no

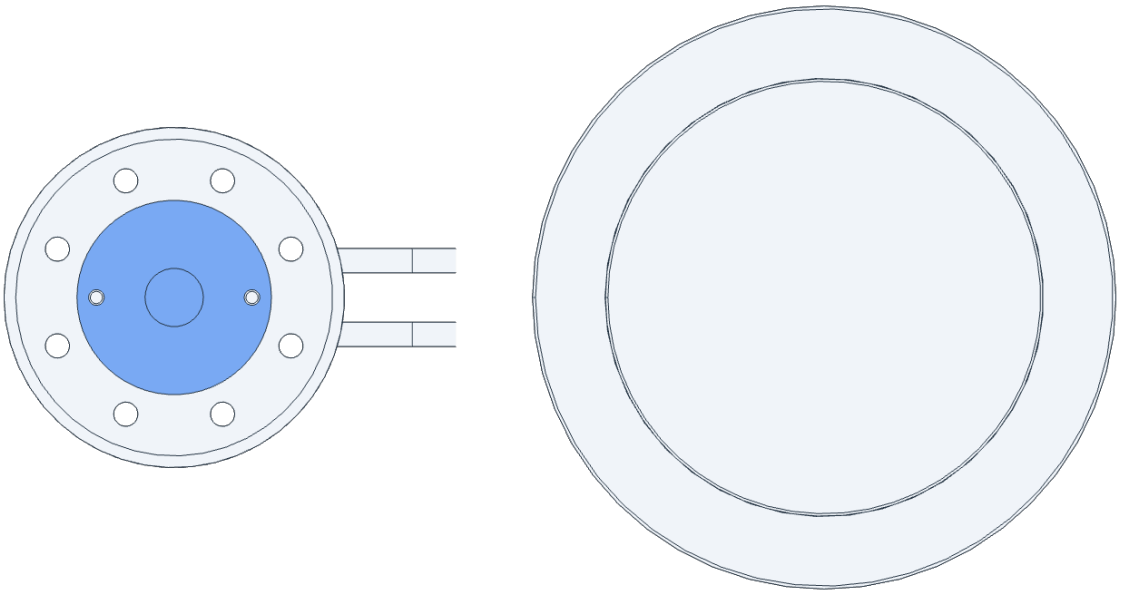
3 Lay the head on the bench, its opening up



You need: ×1 head

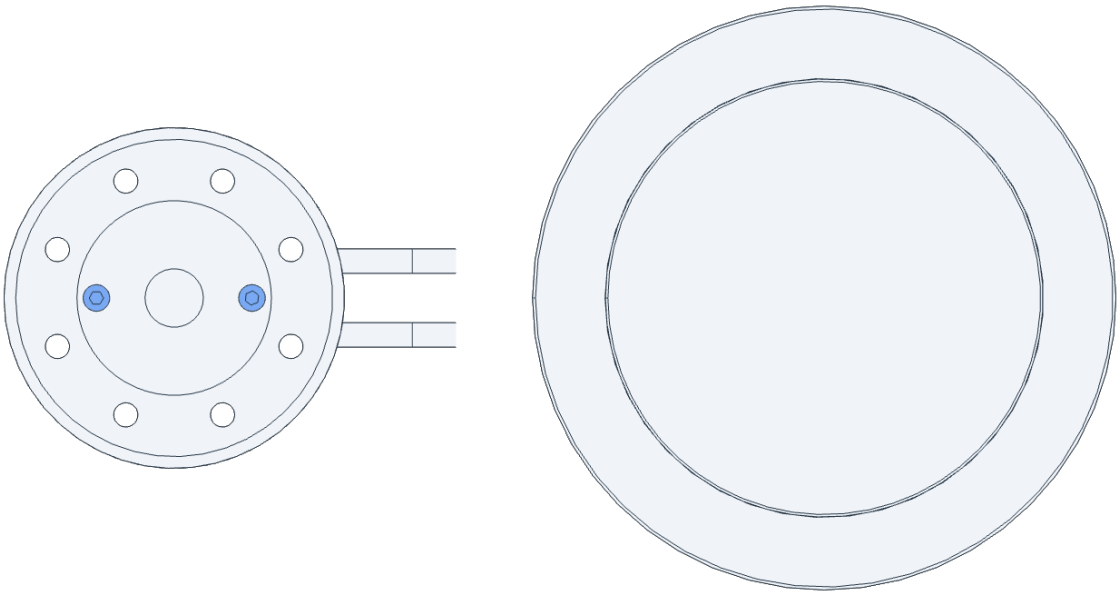
4 Set the LED module on the head's two bosses, its emitter out of the shade

Tip: Lead the LED's wires up through a vent beside the yoke before the board goes on.



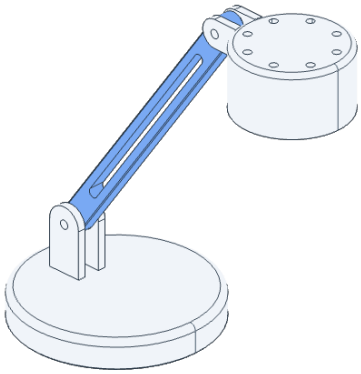
You need: ×1 Led module

5 Screw the LED module to the bosses



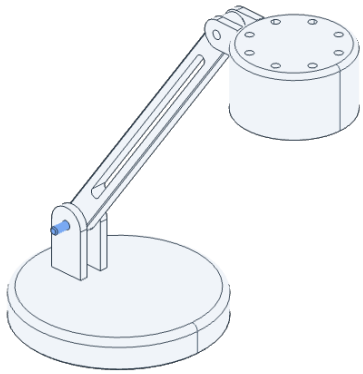
You need: ×1 M3x8 screw:left, ×1 M3x8 screw:right
Hardware: ISO 4762 M3 x 8 socket head screw, qty 2
Torque: not decided yet
The screws cut their own thread in the bosses' 2.5 mm pilot holes: drive them until the board is held, no further.
Where this number comes from: not decided yet

6 Turn the base over and put the arm's lower end between the clevis's cheeks



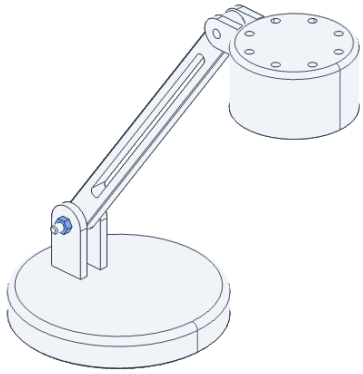
You need: ×1 arm

7 Push the hinge screw through the clevis and the arm, its head on the cheek nearest you



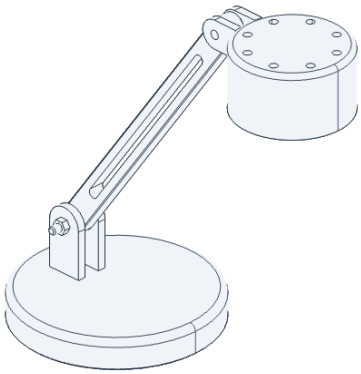
You need: ×1 M5x30 screw:base hinge

8 Run the locknut onto the hinge screw

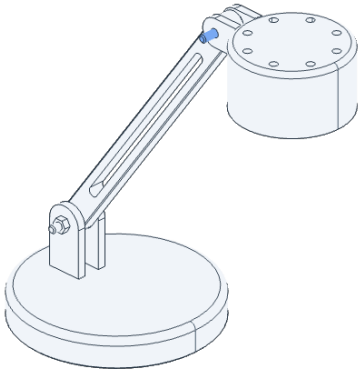


You need: ×1 M5 locknut:base hinge
Hardware: M5 nylon insert locknut on the M5 x 30 hinge screw, qty 1
Torque: not decided yet

9 Put the head's yoke over the arm's upper end, the shade in front

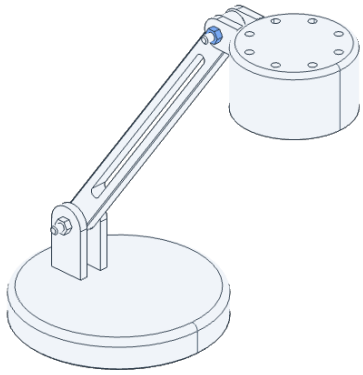


10 Push the hinge screw through the yoke and the arm



You need: ×1 M5x30 screw:head hinge

11 Run the locknut onto the hinge screw

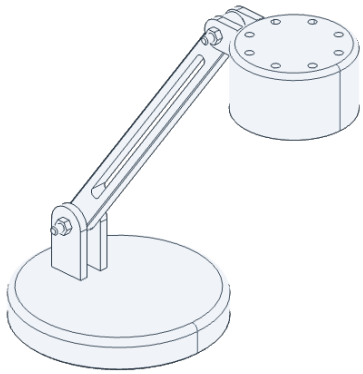


You need: ×1 M5 locknut:head hinge

Hardware: M5 nylon insert locknut on the M5 x 30 hinge screw, qty 1

Torque: not decided yet

12 Tighten each hinge's locknut until the arm and the head hold wherever they are set



Check: With the arm at its lowest, 10 degrees up from the desk, it holds there and the lamp stands
Where this number comes from: not decided yet
Write down: holds, yes or no
Check: With the head square to the arm as assembled, the arm swings from 10 degrees up from the desk, over the top, to 10 degrees below level behind, touching nothing
Where this number comes from: not decided yet
Write down: clear, yes or no
Check: With the arm at 55 degrees up, the head swings from 45 degrees down to 175 degrees up from hanging level, touching nothing
Where this number comes from: not decided yet
Write down: clear, yes or no